



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL SYLLABUS

29-03-2023

## JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE)

### ANSWER KEY

#### PART-1 : PHYSICS

| SECTION-I  | Q. | 1  | 2  | 3  | 4   | 5  | 6  | 7  | 8  | 9  | 10  |
|------------|----|----|----|----|-----|----|----|----|----|----|-----|
|            | A. | D  | D  | C  | C   | C  | C  | A  | C  | D  | B   |
|            | Q. | 11 | 12 | 13 | 14  | 15 | 16 | 17 | 18 | 19 | 20  |
|            | A. | B  | D  | C  | D   | D  | C  | D  | C  | B  | A   |
| SECTION-II | Q. | 1  | 2  | 3  | 4   | 5  | 6  | 7  | 8  | 9  | 10  |
|            | A. | 15 | 0  | 3  | 800 | 10 | 1  | 2  | 4  | 1  | 805 |

#### PART-2 : CHEMISTRY

| SECTION-I  | Q. | 1  | 2  | 3   | 4    | 5   | 6  | 7  | 8  | 9  | 10  |
|------------|----|----|----|-----|------|-----|----|----|----|----|-----|
|            | A. | A  | A  | A   | B    | C   | C  | A  | B  | B  | D   |
|            | Q. | 11 | 12 | 13  | 14   | 15  | 16 | 17 | 18 | 19 | 20  |
|            | A. | C  | D  | A   | A    | C   | B  | D  | A  | C  | A   |
| SECTION-II | Q. | 1  | 2  | 3   | 4    | 5   | 6  | 7  | 8  | 9  | 10  |
|            | A. | 5  | 0  | 246 | 1758 | 504 | 2  | 4  | 14 | 56 | 169 |

#### PART-3 : MATHEMATICS

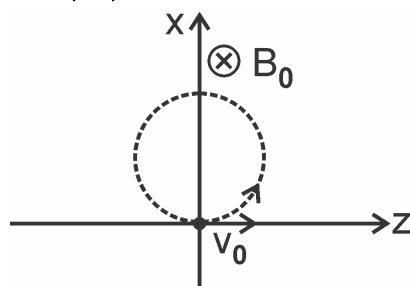
| SECTION-I  | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|------------|----|----|----|----|----|----|----|----|----|----|----|
|            | A. | D  | C  | B  | C  | C  | A  | B  | B  | C  | D  |
|            | Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|            | A. | D  | D  | A  | B  | D  | A  | C  | C  | C  | B  |
| SECTION-II | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|            | A. | 4  | 3  | 2  | 2  | 4  | 0  | 15 | 4  | 11 | 9  |

### HINT – SHEET

#### PART-1 : PHYSICS

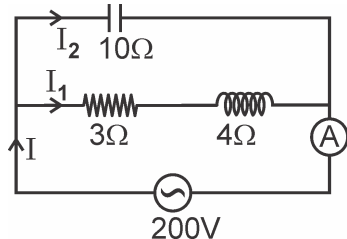
##### SECTION-I

1. Ans (D)



Trajectory of particle is shown in figure.

2. **Ans (D)**



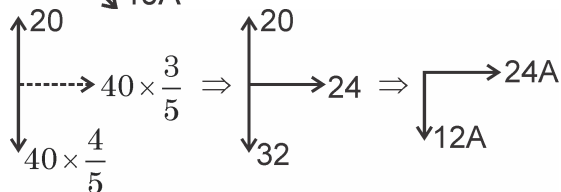
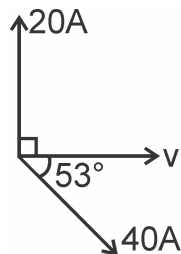
$$I_1 = \frac{200}{\sqrt{3^2 + 4^2}} = 40A$$

$$I_2 = \frac{200}{10} = 20A$$

For L-R Branch

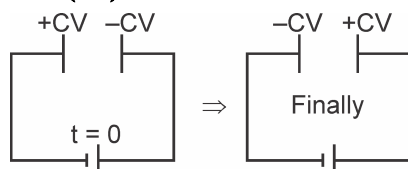
$$\tan \theta = \frac{X_L}{R} = \frac{4}{3}$$

$$\phi = 53^\circ$$



$$\text{Resultant current} = \sqrt{(12)^2 + (24)^2} = 12\sqrt{5}A$$

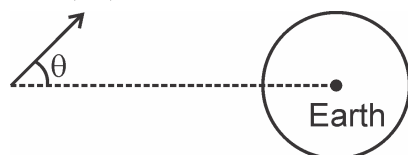
3. **Ans (C)**



$$\text{Work done by battery} = (2 CV)V = 2CV^2$$

Heat loss =  $2CV^2$  as energy stored in capacitor not changes.

4. **Ans (C)**



Angular momentum of object about the centre of earth is non-zero. If it falls on the surface normally then its angular momentum about the centre is zero, but angular momentum is conserved hence it can not hit the surface of earth normally.

5. **Ans (C)**

1  $\rightarrow$  2 Isothermal process;

2  $\rightarrow$  3 adiabatic

3  $\rightarrow$  4 Isothermal process;

4  $\rightarrow$  1 adiabatic

Process 1  $\rightarrow$  2 & 3  $\rightarrow$  4 are isothermal hence

$$A_1 = A_2.$$

6. **Ans (C)**

$$V = 3x^2y; \quad \frac{\partial V}{\partial x} = 6xy; \quad \frac{\partial V}{\partial y} = 3x^2$$

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} = -6xy \hat{i} - 3x^2 \hat{j}$$

$$\vec{E}(1, 1) = -6\hat{i} - 3\hat{j}$$

$$\vec{F} = q\vec{E} = (-2)(-6\hat{j} - 3\hat{j}) = 12\hat{i} + 6\hat{j}$$

$$\vec{a} = 12\hat{i} + 6\hat{j}$$

7. **Ans (A)**

Equation of straight line

$$\frac{dN}{du} = -u + 2$$

$$u_{\text{avg.}} = \frac{\int u dN}{\int dN} = \frac{\int u \frac{dN}{du} \cdot du}{\int \frac{dN}{du} \cdot du}$$

$$u_{\text{avg.}} = \frac{\int u(-4 + 2)du}{\int \text{area under curve}} = \frac{\int_0^2 (-u^2 + 2u)du}{2}$$

$$u_{\text{avg.}} = \frac{\left[-\frac{u^3}{3} + u^2\right]_0^2}{2} = \frac{-\frac{8}{3} + 4}{2} = \frac{2}{3}$$

8. **Ans (C)**

$$u = 10 + (x - 3)^2$$

$$\frac{du}{dx} = 2(x - 3)$$

$$\text{Force } F = -\frac{du}{dx} = -2(x - 3)$$

For mean position  $F = 0 \Rightarrow x = 3$ .

$\therefore$  Minimum potential energy is = 10 J.

Maximum kinetic energy is

$$K_{\text{max.}} = 50 - 10 = 40 \text{ J.}$$

9. **Ans (D)**

Resulting standing wave  $y = y_1 + y_2$

$$y = A \sin\left(\frac{\pi}{4}x - \omega t\right) + A \sin\left(\frac{\pi}{4}x + \omega t + \frac{\pi}{4}\right)$$

$$y = 2A \sin\left(\frac{\pi}{4}x + \frac{\pi}{8}\right) \cos\left(\omega t + \frac{\pi}{8}\right)$$

For position of nodes  $2A \sin\left(\frac{\pi}{4}x + \frac{\pi}{8}\right) = 0$ .

$$\frac{\pi}{4}x + \frac{\pi}{8} = 0, \pi, 2\pi, \dots$$

$$\therefore x = -\frac{1}{2}, \frac{7}{2}, \frac{15}{2}$$

So these nodes are possible in region

$$-1 \leq x \leq 8\text{m}.$$

10. **Ans (B)**

As a tangent on a steady flow gives direction of flow velocity.

$$\text{Slope} = \frac{v_y}{v_x}$$

$$\frac{dy}{dx} = \frac{-x}{y}$$

$$\Rightarrow ydy = -xdx$$

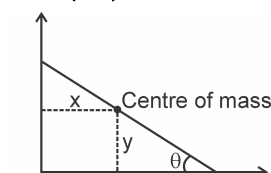
$$\Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + c$$

$$\Rightarrow x^2 + y^2 = 2c$$

11. **Ans (B)**

As pressure on two side of common surface is same so common surface will remain flat in shape.

12. **Ans (D)**



Coordinate of center of mass is

$$x = \frac{L}{2} \cos \theta$$

$$y = \frac{L}{2} \sin \theta$$

$$\therefore x^2 + y^2 = \frac{L^2}{4}$$

Which is equation of circle.

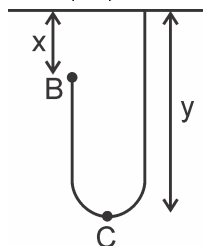
i.e., center of mass moves on a circle of radius  $\frac{L}{2}$ .

$$\therefore \text{Distance travelled} = \frac{2\pi R}{8}$$

$$R = \frac{L}{2} = \frac{3}{4}m$$

$$= \frac{\pi}{4} \times \frac{3}{4} = \frac{3\pi}{16}m$$

13. **Ans (C)**



Length of string is  $L = y + (y - x)$

$$L = 2y - x$$

$$\Rightarrow 0 = 2 \frac{dy}{dt} - \frac{dx}{dt}$$

$$\Rightarrow \frac{2dy}{dt} = \frac{dx}{dt} \Rightarrow 2 \frac{d^2y}{dt^2} = \frac{d^2x}{dt^2}$$

$$\Rightarrow 2a_c = a_0$$

$$\Rightarrow a_c = a_0/2.$$

14. **Ans (D)**

Force constant for string is  $k = \frac{yA}{\ell}$

$$k = \frac{10 \times 10^8 \times (4 \times 10^{-6})}{40 \times 10^{-2}}$$

$$k = 10^4 \text{ N/m}$$

Using energy conservation

$$\frac{1}{2}k(\Delta \ell)^2 = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{k}{m}} \Delta \ell$$

$$v = \sqrt{\frac{10^4}{10 \times 10^{-3}}} \times 4 \times 10^{-2}$$

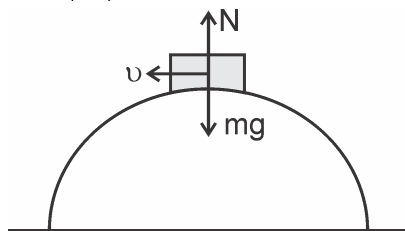
$$v = 40 \text{ m/s}$$

15. **Ans (D)**

$$\epsilon_{eq} = \frac{\frac{\epsilon_1}{r_1} + \frac{\epsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{10}{2} - \frac{6}{1}}{\frac{1}{2} + 1} = \frac{-1}{\frac{3}{2}} = \frac{-2}{3} \text{ volt}$$

As polarity of equivalent battery is -ve so it can not be balanced.

16. Ans (C)



From hemispherical as frame of reference here

$$mg - N = \frac{mv^2}{R}$$

$$\Rightarrow N = mg - \frac{mv^2}{R} \leq 0$$

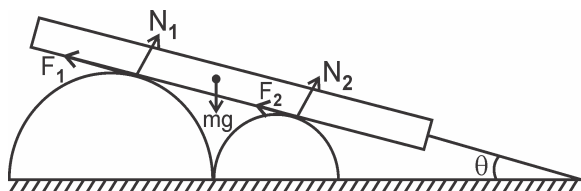
$$\Rightarrow \frac{mv^2}{R} \geq mg$$

$$\Rightarrow v \geq \sqrt{gR}$$

$$\Rightarrow v \geq \sqrt{10}$$

$$\Rightarrow v_{\min.} = \sqrt{10}$$

17. Ans (D)



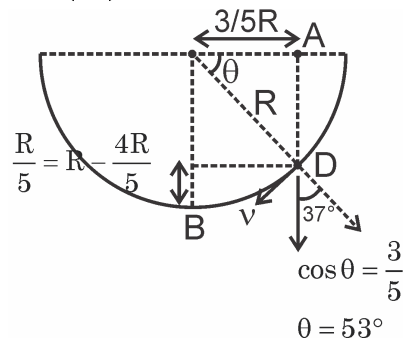
$$\text{In equation } N_1 + N_2 = mg \cos \theta \quad \dots(i)$$

$$F_1 + F_2 = mg \sin \theta$$

$$\mu(mg \cos \theta) = mg \sin \theta$$

$$\tan \theta = \mu$$

18. Ans (C)



As after collision, particle moves along track, so normal component of velocity should be zero. Hence  $e = 0$ .

$$\text{here } AD = \frac{4}{5}R$$

$$\text{For A to D : } mg \left( \frac{4}{5}R \right) = \frac{1}{2}mu^2$$

$u$  is velocity just before striking.

For D to B : using energy conservation

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_B^2 - mg \frac{R}{5}$$

$$\text{here } v = u \sin 37^\circ = \frac{3}{5} \sqrt{\frac{8}{5}gR}$$

$$\Rightarrow v_B^2 = v^2 + \frac{2}{5}gR$$

$$\Rightarrow v_B^2 = \frac{9}{25} \left( \frac{8}{5} \right) gR + \frac{2}{5}gR$$

$$\Rightarrow v_B^2 = \frac{144}{25} + 4 = \frac{244}{25}$$

19. Ans (B)

$$B = \mu_0 n I \text{ inside solenoid}$$

electric field produced in time varying magnetic field is

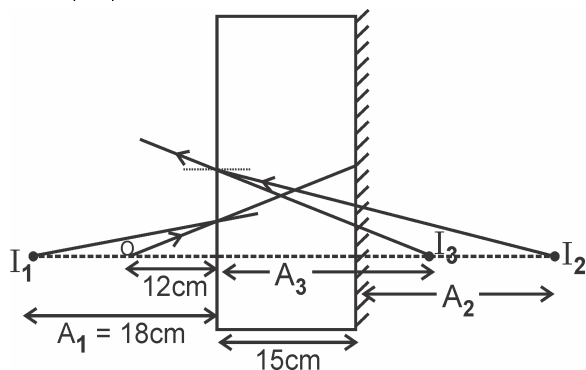
$$E = \frac{r}{2} \frac{dB}{dt}$$

$$E = \frac{r}{2} \mu_0 n \left( \frac{dI}{dt} \right)$$

$$E = \frac{2 \times 10^{-2}}{2} \times \mu_0 \times 50 \times 2$$

$$E = \mu_0$$

20. Ans (A)



$$A_1 = \frac{R}{\mu} = \left(\frac{12}{1}\right) \left(\frac{3}{2}\right) = 18\text{cm}$$

$$A_2 = 18 + 15 = 33\text{ cm} \quad A_3 = \frac{R}{\mu} = \frac{33 + 15}{3/2}$$

$$A_3 = \frac{48}{3} \times 2 = 32\text{cm}$$

∴ Distance of final image  $I_3$  from silvered surface is  $= 32 - 15 = 17\text{ cm}$ .

## PART-1 : PHYSICS

### SECTION-II

1. Ans (15)

6 hr means 3 half life.

After 3 half life no. of active particle  $= \frac{1}{8}$

i.e.,  $\left(1 - \frac{1}{8}\right) = \frac{7}{8}$  will decay.

∴ Probability that substance will decay  $= \frac{7}{8}$ .

2. Ans (0)

As frequency doesn't change hence energy of photon ( $E = hn$ ) remain same.

3. Ans (3)

Energy released  $\epsilon = \Delta mc^2$

$$\epsilon = (7.018 + 1.008 - 2 \times 4.004)(3 \times 10^8)^2 \times 1.67 \times 10^{-27}$$

$$\epsilon = 2.7 \times 10^{-12}\text{ J}$$

Energy released

$$= (2 \times 6.023 \times 10^{23}) \times 2.7 \times 10^{-12}$$

$$= 3.25 \times 10^{12}\text{ J}$$

4. Ans (800)

In the given circuit current is zero in steady state.

Consider potential at B is zero.

$$(V_A - 1200) \times 4 \times 2 + (V_A - 0) \times 2 \times 2 = 0 \text{ (using KCL)}$$

$$V_A = 800\text{ volt.}$$

5. Ans (10)

$$T = 2\pi \sqrt{\frac{I}{MB_H}} = 2\pi \sqrt{\frac{12 \times 10^{-5}}{1600 \times 30 \times 10^{-6}}}$$

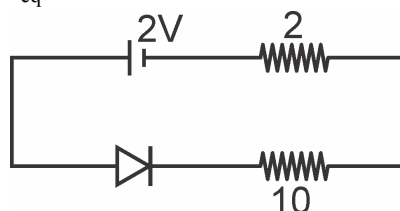
$$T = 2\pi \sqrt{\frac{10^{-2}}{4}}$$

$$T = \frac{\pi}{10}\text{sec.}$$

6. Ans (1)

$$E_{\text{eq.}} = \frac{\frac{12}{3} - \frac{18}{6}}{\frac{1}{3} + \frac{1}{6}} = \frac{4 - 3}{\frac{1}{2}} = 2\text{ volt}$$

$$r_{\text{eq}} = 2\Omega$$



Current through diode

$$I = \frac{2}{12} = \frac{1}{6}\text{A}$$

7. Ans (2)

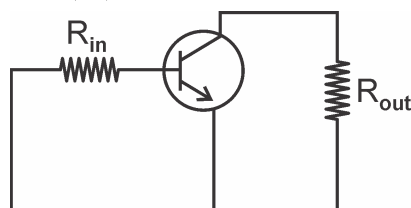
$$\text{Power} = \frac{\text{Energy}}{\text{time}} \Rightarrow p = \frac{\epsilon}{t}$$

$$\text{here } t = \frac{30 \times 10^{-2}}{3 \times 10^8} = 10^{-9}\text{s.}$$

$$\epsilon = Pt = 20 \times 10^{-3} \times 10^{-9}$$

$$= 2 \times 10^{-11}\text{ J}$$

8. Ans (4)



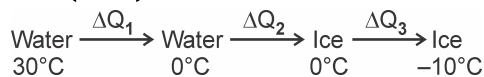
$$\text{Power gain} = \beta^2 \times \frac{R_{\text{out}}}{R_{\text{in}}} = (50)^2 \times \frac{2000}{500} = 10^4$$

9. **Ans (1)**

At point 'P' there is minimum and nearest maxima is at distance  $\beta/2$ .

$$= \frac{\lambda D}{2d} \times \frac{1}{2} = \frac{\lambda D}{4d}$$

10. **Ans (805)**



$$\Delta Q_1 = mS\Delta T = 2000 \times 1 \times 30 = 6 \times 10^4 \text{ cal}$$

$$\Delta Q_2 = mL = 2000 \times 80 = 16 \times 10^4 \text{ cal}$$

$$\Delta Q_3 = mS\Delta T = 2000 \times \frac{1}{2} \times 10 = 10^4 \text{ cal}$$

$$\text{Total heat to be removed} = 23 \times 10^4 \text{ cal}$$

Effective power :

$$p = \frac{\varepsilon}{t} = \frac{23 \times 4.2 \times 10^4}{20 \times 60} = 805 \text{ watt}$$

## PART-2 : CHEMISTRY

### SECTION-I

1. **Ans (A)**

Methemoglobinemia also known as blue baby syndrome is caused by excess of nitrate in drinking water

2. **Ans (A)**

Milk is an oil in water type emulsion

3. **Ans (A)**

$$mvr = \frac{nh}{2\pi}$$

$$-3.4\text{eV} \longrightarrow n = 2$$

$$-13.6\text{ eV} \longrightarrow n = 1$$

4. **Ans (B)**

- (A) Synthetic detergents contains no soap  
(B) Anionic detergents are used in toothpastes  
(C) Liquid dish washers use anionic detergents  
(D) Synthetic detergents with branched chains can be biodegraded with difficulty

5. **Ans (C)**

Meso compound is optically inactive even though it contains more than one chiral centres.

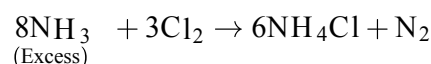
6. **Ans (C)**

$$X_{8 \times \frac{1}{8} + 6 \times \frac{1}{2}} Y_{1 + 8 \times \frac{1}{4}} \\ = X_4 Y_3$$

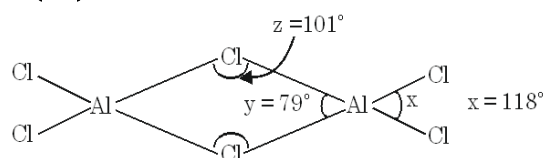
7. **Ans (A)**

$[\text{Co(en)}_3] \text{Cl}_3$  has no geometrical isomers but shows optical activity due to lack of plane of symmetry.

8. **Ans (B)**

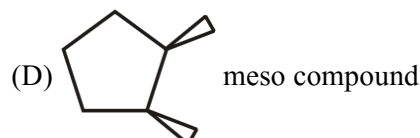
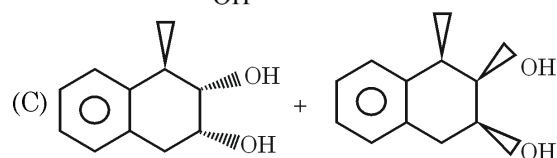
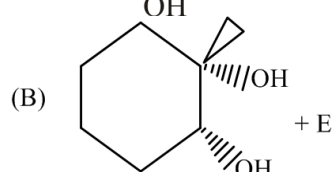
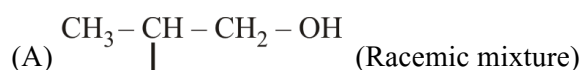


9. **Ans (B)**

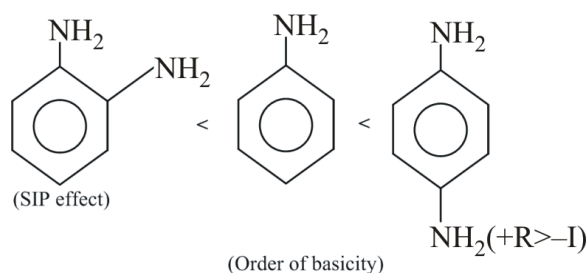


10. **Ans (D)**

Products



11. **Ans (C)**



12. **Ans (D)**

Natural rubber is all trans configuration

13. **Ans (A)**

In ellingham diagram lower metal can reduce higher metal oxide

14. **Ans (A)**

Solution

(A) Rate of hydrolysis  $\propto$  lewis acidity

(B) Rate of hydrolysis  $\propto$  oxidation state of central atom

(C) Rate of hydrolysis  $\propto$  lewis acidity

$\text{SiF}_4 > \text{SiCl}_4 > \text{SiBr}_4 > \text{SiI}_4$

(D) as size of central atom increases, steric crowding decreases and rate of hydrolysis increase

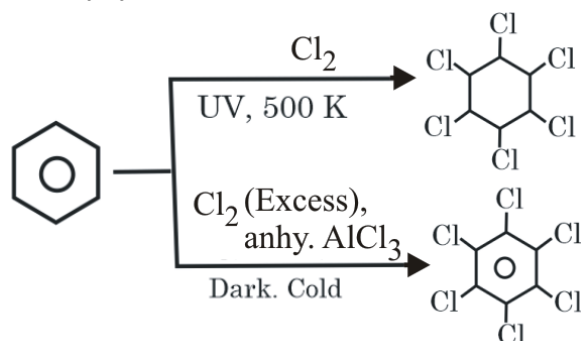
15. **Ans (C)**

A = CaO

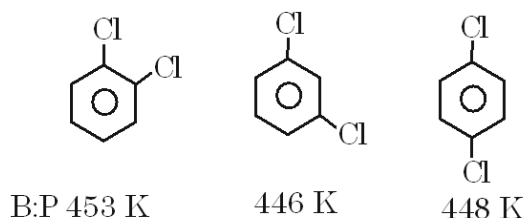
B =  $\text{CaC}_2$

C =  $\text{C}_2\text{H}_2$

D =  $\text{C}_6\text{H}_6$

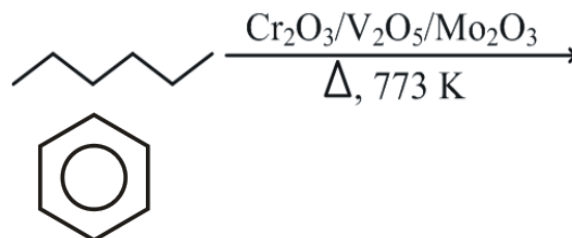
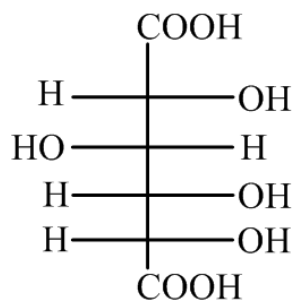


16. **Ans (B)**



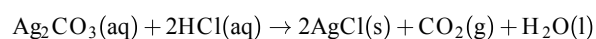
17. **Ans (D)**

P =  $\text{HNO}_3$       R = n - Hexane

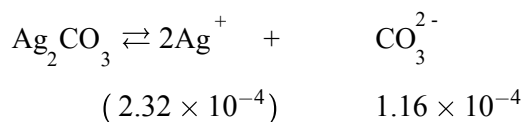


**Benzene**

18. **Ans (A)**



The moles of  $\text{CO}_2$  produced will be equal to  $\text{CO}_3^{2-}$  in  $\text{Ag}_2\text{CO}_3$  due to stoichiometric equation



$$K_{\text{sp}} = (2.32 \times 10^{-4})^2 (1.16 \times 10^{-4})$$

$$K_{\text{sp}} = 6.2 \times 10^{-12}$$

19. **Ans (C)**

$\text{Li}_2\text{O}$  and  $\text{MgO}$  do not form superoxides

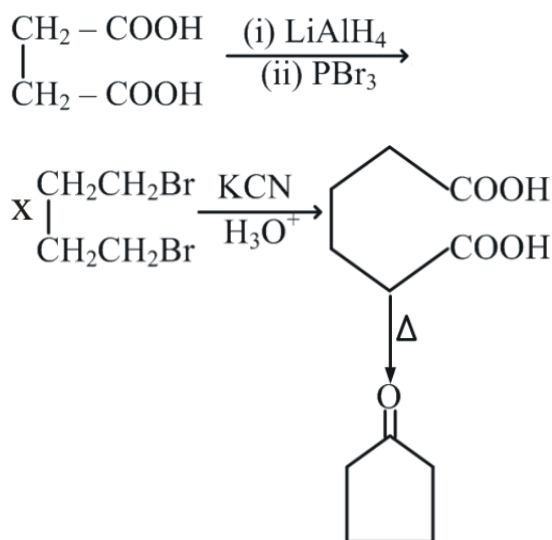
20. **Ans (A)**

Based on inter molecular forces

## PART-2 : CHEMISTRY

### SECTION-II

1. **Ans ( 5 )**



2. **Ans ( 0 )**

$$n_{\text{ice}} = \frac{74.6}{18} = 4.14$$

$$\Delta S_{\text{sys}} = \frac{4.14 \times 6010}{273} = 91.1 \text{ J/K} - \text{mol}$$

$$\Delta S_{\text{surr}} = \frac{-(4.14 \times 6010)}{273} = -91.1 \text{ J/mol} - \text{K}$$

$$\Delta S_{\text{univ}} = 0$$

3. **Ans ( 246 )**

Amount of charge that 4.0 g of  $\text{MnO}_2$  can produce

$$4 \times \frac{1}{87} \times \frac{2}{2} \times 96500 = 4.44 \times 10^3 \text{ C}$$

$$0.005 \text{ A} = 0.005 \text{ C/s}$$

$$\therefore 4.44 \times 10^3 \times \frac{1}{0.005} \times \frac{1}{3600} = 2.46 \times 10^2 \text{ hr}$$

4. **Ans ( 1758 )**

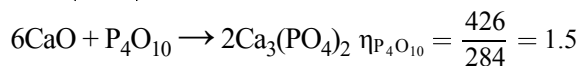
$$K_{\text{cat}} = K_{\text{uncat}}$$

$$\therefore \frac{E_{\text{a(cat)}}}{T_1} = \frac{E_{\text{a(uncat)}}}{T_2}$$

$$\frac{7}{293} = \frac{42}{T_2}$$

$$\therefore T_2 = 1.758 \times 10^3 \text{ K}$$

5. **Ans ( 504 )**



No. of moles of CaO required =  $1.5 \times 6 = 9$

mass of CaO required =  $56 \times 9 = 504 \text{ g}$

6. **Ans ( 2 )**

$$a - 2r = 1.2 \text{ \AA} \quad \text{and} \quad \sqrt{3} a = 4r$$

$$a - 2 \times \frac{\sqrt{3}}{4} a = 1.2 \text{ \AA}$$

$$a \left( 1 - \frac{\sqrt{3}}{2} \right) = 1.2 \text{ \AA}$$

$$a = 8 \text{ \AA}$$

7. **Ans ( 4 )**

$\text{Zn}^{2+}(\text{aq})$ ,  $\text{Ca}^{2+}(\text{aq})$ ,  $\text{Sc}^{3+}(\text{aq})$ ,  $\text{Pb}^{2+}(\text{aq})$  are colorless due to absence of unpaired electrons

8. **Ans ( 14 )**

Morphine is  $\text{C}_{17}\text{H}_{19}\text{NO}_3$

codeine is  $\text{C}_{18}\text{H}_{21}\text{NO}_3$

9. **Ans ( 56 )**

$$\%N = \frac{1.4 \times 2.5 \times 2 \times 2}{0.25} = 56$$

10. **Ans ( 169 )**

This is an example of reverse osmosis.

$$\pi = MRT = 0.7 \times 0.0821 \times 293 = 16.8 \text{ atm}$$

The water pressure at the membrane depends on the height of the sea above it, i.e. the depth  $P = \rho gh$ , and fresh water will begin to pass through the membrane when  $P = \pi$ .

$$h = \frac{\pi}{\rho g} = \frac{1.70 \times 10^6 \text{ Pa}}{9.81 \times 1.03 \times 10^3} = 169 \text{ m}$$

## PART-3 : MATHEMATICS

### SECTION-I

#### 1. Ans (D)

$$n(s) = 9 \times 10 \times 10 \times 10 = 9000$$

Let abcd be a 4 digit number in AP then

$$d = 0, \{(1111), (2222), \dots, (9999)\} = 9$$

$$d = 1 \quad (1, 2, 3, 4), (2, 3, 4, 5), \dots, (6, 7, 8, 9) = (6)$$

But  $d = -1$  six cases in reverse order + 1

case  $\{3210\}$  as 0 can be in last place = 7

$$d = 2 \quad \{1, 3, 5, 7\} \quad \{2, 4, 6, 8\} \quad \{3, 5, 7, 9\} = (3)$$

$$d = -2, \text{ Three case above in reverse } \{6, 4, 2, 0\} = 4$$

$d = 3$  not possible

$d = -3$  one case 9630 is possible = 1

$$n(A) = 30$$

$$P(A) = \frac{30}{9000} = \frac{1}{300}$$

#### 2. Ans (C)

The radius of small circle is maximum, when the centre of circles should lie on line joining the vertices of larger triangle with centre of larger circle

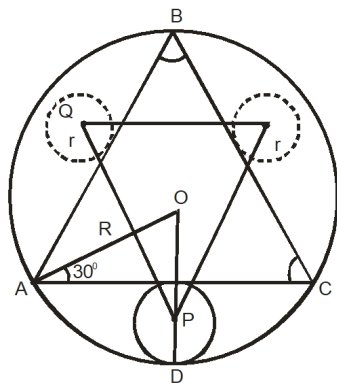
$$R = OA = 8\text{cm}$$

$$OP = 8 \sin 30^\circ + r = 4 + r$$

$$OD = 4 + 2r = 8 \Rightarrow r = 2$$

$$\text{In } \triangle PQR, PQ = 2 OP \cos 30^\circ$$

$$\Rightarrow PQ = 6 \cdot \frac{\sqrt{3}}{2} \times 2 = 6\sqrt{3}$$



PQR is also an equilateral triangle

$$\text{ar } (\triangle PQR) = \frac{\sqrt{3}}{4} \times (6\sqrt{3})^2 = 27\sqrt{3} \text{ cm}^2$$

#### 3. Ans (B)

$$\text{let } g(x) = f^{-1}(x) = x^3 + 3x - 4$$

$$\Rightarrow f(g(x)) = x$$

$$\Rightarrow f'(g(x)) = \frac{1}{g'(x)}$$

Differentiating again

$$f''(g(x)) \cdot g'(x) = \frac{-1}{(g'(x))^2} \cdot g''(x)$$

$$f''(g(x)) = \frac{-g''(x)}{(g'(x))^3}$$

For  $f''(0)$ , put

$$g(x) = 0 \Rightarrow x^3 + 3x - 4 = 0 \Rightarrow x = 1$$

$$\text{So, } f''(0) = \frac{-g''(1)}{(g'(1))^3}$$

$$\text{Now } g(x) = x^3 + 3x - 4$$

$$g'(x) = 3x^2 + 3, g'(1) = 6$$

$$g''(x) = 6x, g''(1) = 6$$

$$\text{So, from, } f''(0) = \frac{-6}{(6)^3} = \frac{-1}{36}$$

#### 4. Ans (C)

$$a = \lim_{x \rightarrow 1} \frac{9}{1-x^9} - \frac{7}{1-x^7} \dots \dots \dots (1)$$

$$\text{replace } x \rightarrow \frac{1}{x},$$

$$a = \lim_{x \rightarrow 1} \left( \frac{9}{1-\frac{1}{x^9}} - \frac{7}{1-\frac{1}{x^7}} \right) = \lim_{x \rightarrow 1} \left( \frac{-9x^9}{1-x^9} + \frac{7x^7}{1-x^7} \right)$$

$$\dots \dots (2)$$

add (1) & (2)

$$2a = \lim_{x \rightarrow 1} \left( \frac{9(1-x^9)}{1-x^9} - \frac{7(1-x^7)}{1-x^7} \right) = 2$$

$$a = 1,$$

$$\text{Now, } r = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\ln^3(1+x)}, \text{ as}$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

$$= \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x(1 - \cos x)}{x \cdot x^2} = \frac{1}{2}$$

$$\text{So, } S = \frac{a}{1-r} = \frac{1}{1-1/2} = 2$$

$$\text{Now for hyperbola, } \frac{x^2}{1} - \frac{y^2}{3} = 1$$

$$e = \sqrt{1 + \frac{(\sqrt{3})^2}{1^2}} = 2$$

5. Ans (C)

- Let P (x, y) be any point on curve

equation of tangent at P

$$Y - y = m (X - x)$$

$$\left| \frac{y - mx}{\sqrt{1 + m^2}} \right| = x \text{ Squaring both side}$$

$$\Rightarrow y^2 - x^2 = m (2xy)$$

$$\Rightarrow \frac{y^2 - x^2}{2xy} = \frac{dy}{dx} \left( \because m = \frac{dy}{dx} \right)$$

Homogeneous

Put  $y = vx$

$$\therefore v \times \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = - \frac{v^2 - 1}{2v}$$

$$\Rightarrow \int \frac{2v}{v^2 + 1} dv = - \int \frac{dx}{x}$$

$$\Rightarrow \ln |v^2 + 1| = -\ln x + \ln c$$

$$\Rightarrow \frac{y^2}{x^2} + 1 = \frac{c}{x}$$

(1, 2) satisfies Eq<sup>n</sup>

$$\Rightarrow \frac{4}{1} + 1 = \frac{C}{1}, C = 5$$

$$\therefore \frac{y^2 + x^2}{x^2} = \frac{5}{x}$$

$$\Rightarrow x^2 + y^2 - 5x = 0$$

Option C is correct

6. Ans (A)

$$I_3 = \int_0^{\pi/2} \cos(\pi \sin^2 x) dx \dots \dots (1)$$

$$I_3 = \int_0^{\pi/2} \cos(\pi \cos^2 x) dx \dots \dots (2)$$

adding equation 1 and 2

$$2I_3 = \int_0^{\pi/2} \cos(\pi \cos^2 x) + \cos(\pi \sin^2 x) dx$$

$$2I_3 = \int_0^{\pi/2} 2 \cos(\pi/2) + \cos\left(\frac{\pi}{2} \cos 2x\right) dx$$

$$I_3 = 0$$

$$I_2 = \int_0^{\pi/2} \cos(\pi (1 - \cos 2x)) dx$$

$$= - \int_0^{\pi/2} \cos(\pi \cos 2x) dx$$

$$2x = t, 2 = \frac{dt}{dx}$$

$$I_2 = -\frac{1}{2} \int_0^{\pi} \cos(\pi \cos t) dt$$

$$I_2 = -\frac{2}{2} \int_0^{\pi/2} \cos(\pi \cos t) dt$$

$$I_2 = - \int_0^{\pi/2} \cos(\pi \cos t) dt$$

$$\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$$

Put  $t = x$

If  $f(2a - x) = f(x)$

$$I_2 = - \int_0^{\pi/2} \cos(\pi \cos x) dx$$

$$- \int_0^{\pi/2} \cos(\pi \sin x) dx = -I_1$$

$$\Rightarrow I_1 + I_2 = 0$$

(A) is incorrect

7. **Ans (B)**

roots of equation are

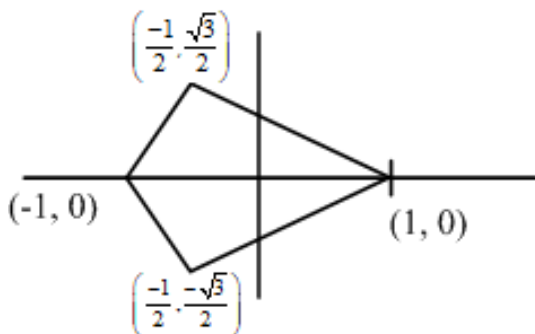
$$1, -1, w \text{ and } w^2$$

$$\downarrow \quad \downarrow$$

$$\frac{-1 + \sqrt{3}i}{2} \quad \frac{-1 - \sqrt{3}i}{2}$$

So vertices are

$$(1, 0), (-1, 0), \left(\frac{-1}{2}, \frac{\sqrt{3}}{2}\right) \text{ and } \left(\frac{-1}{2}, \frac{-\sqrt{3}}{2}\right)$$



Area = 2 x

$$\frac{1}{2} \begin{vmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1 & 0 \\ 1 & 0 \end{vmatrix} = \left| \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right| = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}$$

8. **Ans (B)**

$$3a^2 + b^2 + c^2 + 4ab + 2ca = 0$$

$$4a^2 + 4ab + b^2 = a^2 + c^2 - 2ac$$

$$(2a + b)^2 = (a - c)^2$$

$$2a + b = a - c \text{ or } 2a + b = -a + c$$

$$a + b + c = 0 \text{ or } 3a + b - c = 0$$

as  $ax + by + c = 0$  is family of lines are concurrent therefore points are (1,1) and (-3, -1).

Now circle with least radius i.e. fixed point will be diametric end point so equation of circle will be

$$(x - 1)(x + 3) + (y - 1)(y + 1) = 0$$

9. **Ans (C)**

Let points on hyperbola be

$$(x_1, y_1), (x_2, y_2), (x_3, y_3) \text{ and } (x_4, y_4)$$

and as  $x_1 y_1 = a^2$  so  $y_1 = \frac{a^2}{x_1}$  and similarly

$$y_2 = \frac{a^2}{x_2}, y_3 = \frac{a^2}{x_3} \text{ and } y_4 = \frac{a^2}{x_4}$$

Now  $(x_1, y_1) = \left(x_1, \frac{a^2}{x_1}\right)$  also satisfy circle

$$\text{So, } x_1^2 + \frac{a^4}{x_1^2} + 2gx_1 + 2f\frac{a^2}{x_1} + c = 0$$

Similarly for  $(x_2, y_2), (x_3, y_3)$  and  $(x_4, y_4)$

also satisfy the equation

$$x^4 + 2gx^3 + cx^2 + 2fa^2x + a^4 = 0 \text{ has roots}$$

$$x_1, x_2, x_3, x_4 \text{ so } x_1 + x_2 + x_3 + x_4 = -2g,$$

$$x_1 x_2 x_3 x_4 = a^4$$

$$\text{Now } y_3 y_4 = \frac{a^2}{x_3} \cdot \frac{a^2}{x_4} = \frac{a^4}{x_3 x_4} (x_1 x_2)$$

$$\text{Hence } x_1 x_2 = y_3 y_4$$

10. **Ans (D)**

$$\begin{vmatrix} 1 & -1 & -1 \\ 1 & -1 & \sin \theta \\ 1 & \sin \theta & -1 \end{vmatrix} = 0$$

$$\Rightarrow 1(1 - \sin^2 \theta) + 1(-1 - \sin \theta) - 1(\sin \theta + 1) = 0$$

$$\Rightarrow 1 - \sin^2 \theta - 1 - \sin \theta - \sin \theta - 1 = 0$$

$$\Rightarrow \sin^2 \theta + 2 \sin \theta + 1 = 0$$

$$\Rightarrow (\sin \theta + 1)^2 = 0$$

$$\sin \theta = -1$$

$$\theta = (4n - 1) \frac{\pi}{2}, n \in \mathbb{I}$$

$$\theta = \frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}, \frac{15\pi}{2}, \dots$$

So 1011 Solutions possible

11. Ans (D)

Let  $\sqrt{4-x^2} = t, x \in [-2, 2], t \in [0, 2]$

$$f(t) = (3-t)^2 + (1+t)^3$$

$$f(t) = 2(3-t)(-1) + 3(1+t)^2 = 0$$

$$\Rightarrow 3(1+t)^2 = 2(3-t)$$

$$\Rightarrow 3t^2 + 6t + 3 = 6 - 2t$$

$$\Rightarrow 3t^2 + 8t - 3 = 0$$

$$\Rightarrow 3t^2 + 9t - t - 3 = 0$$

$$\Rightarrow 3t(t+3) - 1(t+3) = 0$$

$$\Rightarrow (t+3)(3t-1) = 0$$

$$t = -3, \frac{1}{3}$$

As  $t \in [0, 2]$  so  $t = 1/3$

$$f(1/3) = (3-1/3)^3 = \frac{64}{9} + \frac{64}{27} = \frac{192+64}{27} = \frac{256}{27}$$

So range of function is  $\left[\frac{256}{27}, 28\right]$

$$\text{So } P - Q = 28 - \frac{256}{27} = \frac{500}{27}$$

12. Ans (D)

$$(a) \quad 3x + y - z = 0 \dots\dots\dots(1)$$

$$x - \frac{py}{4} + z = 0 \dots\dots\dots(2)$$

$$2x - y + 2z = q \dots\dots\dots(3)$$

$$D = \begin{vmatrix} 3 & 1 & -1 \\ 1 & -\frac{p}{4} & 1 \\ 2 & -1 & 2 \end{vmatrix} = 4 - 2p$$

$$D_1 = \begin{vmatrix} 0 & 1 & -1 \\ 2 & -\frac{p}{4} & 1 \\ q & -1 & 2 \end{vmatrix} = q - 2 - \frac{pq}{4}$$

$$D_2 = \begin{vmatrix} 3 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & q & 2 \end{vmatrix} = 16 - 4q$$

$$D_3 = \begin{vmatrix} 2 & q & 2 \\ 3 & 1 & 0 \\ 1 & -\frac{p}{4} & 2 \end{vmatrix} = -q \left( \frac{3p}{4} + 1 \right) + 10$$

Unique solution,  $D \neq 0$

$p \neq 2, q \in \mathbb{N}$

$\Rightarrow$  Number of ordered pairs  $(p, q)$  in  $[1, 10]$  are 90

13. Ans (A)

Required plane will contain the points  $(1, 2, 3)$  and

$(-1, -3, 1)$  and parallel to line having direction ratios

$(2, 3, 4)$ .

let equation of plane be  $a(x-1) + b(y-2) + c(z-3) = 0$

$$= 0$$

Now  $(-1, -3, 1)$  also satisfy above equation

$$a(-1-1) + b(-3-2) + c(1-3) = 0$$

$$\Rightarrow -2a - 5b - 2c = 0 \dots\dots\dots(1)$$

$$\text{and } 2a + 3b + 4c = 0 \dots\dots\dots(2)$$

solving (1) and (2)

$$\frac{a}{-20+6} = \frac{b}{-4+8} = \frac{c}{-6+10}$$

$$\Rightarrow \frac{a}{-7} = \frac{b}{2} = \frac{c}{2} = k(\text{let})$$

so,  $a = -7k, b = 2k, c = 2k$

therefore equation of plane is

$$-7k(x-1) + 2k(y-2) + 2k(z-3) = 0$$

$$\Rightarrow 7(x-1) - 2(y-2) - 2(z-3) = 0$$

$$\Rightarrow 7x - 2y - 2z + 3 = 0$$

14. Ans (B)

For  $S : (\sim S_1 \wedge S_2) \rightarrow S_3$

$\sim (S_1 \vee \sim S_2) \rightarrow S_3$

15. Ans (D)

- No of ways of selecting one card from

Ashok's bag and other from shilpa's bag

$$= {}^{40}C_1 \times {}^5C_1 = 200$$

Now, if card taken from shilpa's bag shows 1

than 1 will divide all numbers of Ashok's card.

[40 ways]

If card is 2, then Remainder is 0 or 1

= [40 ways]

If card is 3 then also 40 ways

But, If card from shilpa's bag is 4 then

$(4n + 3)$  will be rejected

i.e., 10 ways, number's left =  $40 - 10 = 30$

and card shows 5 then,  $5n + 3, 5n + 4$

numbers will be rejected, Total no

$$= \frac{39 - 3}{4} + 1 = 10$$

Numbers left =  $40 - 10 = 30$

$$\text{Numbers of } (5n + 3) \text{ form} = \frac{39 - 4}{5} + 1 = 8$$

$$\text{Numbers of } (5n + 4) \text{ form} = \frac{38 - 3}{5} + 1 = 8$$

Total numbers left =  $40 - 10 - 8 - 8 = 24$

$$P(E) = \frac{(40 \times 3) + (30) + 24}{200} = \frac{174}{200}$$

Ans. 0.87

16. Ans (A)

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$$

Now,  $A^2 = 0$

$$A^3 = A^4 = \dots = A^{50} = 0$$

$$\text{Now, } (I + A)^{50} - 50A = I + {}^{50}C_1 A + {}^{50}C_2 A^2$$

$$+ \dots + {}^{50}C_{50} A^{50} - 50A = I + 50A - 50A = I$$

$$\det((I + A)^{50} - 50A) = 1$$

17. Ans (C)

let  $\vec{a}, \vec{b}, \vec{c}$  and  $\vec{d}$  be the position vectors of four equally inclined vectors coterminous at origin. so any three out of four vectors form an equilateral triangle hence these four vectors form regular tetrahedron.

$$\vec{d} = -\vec{a} - \vec{b} - \vec{c}$$

$$\vec{d} \cdot \vec{d} = |\vec{a} + \vec{b} + \vec{c}|^2$$

Let angle is  $\alpha$

$$1 = 1 + 1 + 1 + 2(1 + 1 + 1) \cos \alpha$$

$$\Rightarrow \cos \alpha = \frac{-1}{3}$$

$$\sin \alpha = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow a^2 + b^2 = 13$$

18. Ans (C)

$$\text{let } P(x) = x^4 + ax^3 + bx^2 + cx + d$$

and roots of

$$P(x) = 0 \text{ are } \alpha, \beta, \gamma \text{ and } \delta$$

so,  $a, b, c, d, \in I$

assuming  $\alpha \leq \beta \leq \gamma \leq \delta$

So if  $\gamma + \delta = 10, \delta + \beta = 9, \alpha + \beta = 1,$   
 $\alpha + \gamma = 2,$

Now for  $\alpha + \delta$  and  $\beta + \gamma$  we may consider two cases

case (1)  $\alpha + \delta = 5$  and  $\beta + \gamma = 6$

Now solving  $\alpha = \frac{-3}{2}, \beta = \frac{5}{2}, \gamma = \frac{7}{2}$  and

$\delta = \frac{13}{2}$  It is not possible as product of roots

i.e. d not integer

case (2)  $\alpha + \delta = 6$  and  $\beta + \gamma = 5$

So,  $\alpha = -1, \beta = 2, \gamma = 3, \delta = 7$

$$\text{So, } P(x) = (x+1)(x-2)(x-3)(x-7)$$

$$P\left(\frac{1}{2}\right) = \left(\frac{-3}{2}\right)\left(\frac{-5}{2}\right)\left(\frac{-13}{2}\right) = \frac{-585}{16}$$

19. Ans (C)

$$\text{Put } \int_0^{\pi/2} \theta \sin^8 \theta d\theta = I, \int_0^{\pi/2} \theta \sin^{10} \theta d\theta = P$$

$$P = \int_0^{\pi/2} (1 - \cos^2 \theta) \sin^8 \theta d\theta = I, - \int_0^{\pi/2} \theta \cos^2 \theta (\sin^8 \theta) d\theta$$

$$= I - \int_0^{\pi/2} (\theta \cos \theta) (\cos \theta \sin^8 \theta) d\theta.$$

Using integration by parts we get

$$P = I - \left[ \frac{(\theta \cos \theta) \sin^9 \theta}{9} - \int \left( \frac{-\theta \sin \theta + \cos \theta}{9} \right) \sin^9 \theta d\theta \right]_0^{\pi/2}$$

$$P = I - \left( \frac{P}{9} \right) + \frac{1}{9} \int_0^{\pi/2} \sin^9 \theta \cos \theta d\theta, \frac{10P}{9} = I + \frac{1}{90}$$

$$\therefore 100P = 90I + 1 \therefore \frac{100P - 1}{I} = 90$$

20. Ans (B)

$$\text{put } xe^x = t$$

$$(e^x + xe^x)dx = dt$$

$$\therefore \frac{dt}{dy} + (ye^y - t) = 0$$

$$\Rightarrow \frac{dt}{dy} - t + ye^y = 0$$

$$\text{I.F. } e^{-\int dy} = e^{-y}$$

$$t \cdot e^{-y} = - \int ye^y e^{-y} dy$$

$$x e^x e^{-y} = - \frac{y^2}{2} + C$$

$$f(0) = 0$$

$$\Rightarrow C = 0; 2x e^x e^{-y} + y^2 = 0$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (4)

using concept of differentiation of determinant

$$f'(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(\beta-\gamma) & \cos(\gamma-\alpha) & \cos(\alpha-\beta) \end{vmatrix} +$$

$$\begin{vmatrix} \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ -\sin(x+\alpha) & -\sin(x+\beta) & -\sin(x+\gamma) \\ \cos(\beta-\gamma) & \cos(\gamma-\alpha) & \cos(\alpha-\beta) \end{vmatrix}$$

$$+ \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ 0 & 0 & 0 \end{vmatrix}$$

$$\text{So, } f'(x) = 0$$

$$\Rightarrow f(x) = \text{constant as } f(100) = k \text{ so}$$

$$\sum_{r=1}^{100} f(r) = 100k$$

2. Ans (3)

Let students have  $x_1, x_2, x_3, x_4, x_5$  in increasing order of their weights

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = 40 \quad (\text{Given})$$

$$\frac{x_2 + x_3 + x_4 + x_5}{4} = 45 \quad (\text{Given})$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 160 + x_5,$$

Sum of weights of all the students)

$$x_2 + x_3 + x_4 + x_1 + x_5 = 180 + x_1,$$

(Sum of weights of all the students)

$$160 + x_5 = 180 + x_1 \quad (1)$$

Average maximum  $\Rightarrow x_5$  is maximum

$$x_5 - x_1 = 20, x_5 = x_1 + 20$$

$$(x_5)_{\max} = (x_1)_{\max} + 20 = 40 + 20 = 60$$

$$\text{Average maximum} = 220/5 = 44$$

now, average is minimum when least weight is minimum

$$x_1 = x_5 - 20 \text{ min} \rightarrow \text{Avg g heaviest (45)}$$

$$\therefore (\bar{x})_{\min} = \frac{205}{5} = 41$$

$$\therefore (\bar{x})_{\max} - (\bar{x})_{\min} = 44 - 41 = 3 \text{ kg}$$

3. **Ans (2)**

$$\text{Let } F = x^3y + xy^3 \\ = xy(x^2 + y^2)$$

$$\text{Let } x = r \cos \theta, y = r \sin \theta, x^2 + y^2 = r^2 \text{ so}$$

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta + r^2 \cos \theta \sin \theta = 1$$

$$\Rightarrow r^2 = \frac{1}{1 + \sin \theta \cos \theta} = \frac{2}{2 + \sin 2\theta}$$

$$\sin 2\theta = \frac{2}{r^2} - 2$$

$$\text{As } -1 \leq \sin 2\theta \leq 1$$

$$-1 \leq \frac{2}{r^2} - 2 \leq 1$$

$$1 \leq \frac{2}{r^2} \leq 3$$

$$\frac{1}{2} \leq \frac{1}{r^2} \leq \frac{3}{2}$$

$$\frac{2}{3} \leq r^2 \leq 2$$

$$\text{Now } F = (1 - r^2)r^2 \text{ let } r^2 = t, t \in \left[\frac{2}{3}, 2\right]$$

$$F = (1 - t)t \Rightarrow \frac{dF}{dt} = 1 - 2t \forall t \in \left[\frac{2}{3}, 2\right]$$

$$\text{So } F \in \left[-2, \frac{2}{9}\right] \text{ so } |F| \leq 2$$

4. **Ans (2)**

We have

$$\log_{a^2} |x + y| = \frac{1}{2} \Rightarrow |x + y| = a \dots (1)$$

$$\log_a \left( \frac{y}{|x|} \right) = \frac{1}{2} \log_a 4 \Rightarrow y = 2|x| \dots (2)$$

Case-I, If  $x > 0$ , then  $y = 2x$ ,

$$\Rightarrow |3x| = a, x = a/3, y = 2a/3$$

Case-II, If  $x < 0$ , then  $y = -2x$

$$\Rightarrow |-x| = a$$

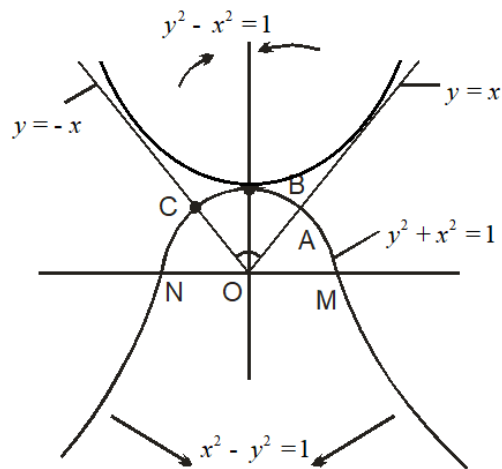
$$\Rightarrow x = -a (\text{since } a > 0), y = 2a$$

two ordered pairs

5. **Ans (4)**

In 1<sup>st</sup> and 2<sup>nd</sup> Quadrants we have

$$x^2 + y^2 = 1 \text{ and } y^2 - x^2 = 1$$



In 3<sup>rd</sup> & 4<sup>th</sup> Quadrant

$$x^2 - y^2 = 1$$

$$\text{Area (OABO)} = \frac{1}{2} \text{ as } (\text{OMABO}) = \frac{\pi}{4} \times \frac{1}{2} = \pi/8$$

Similarly as

$$(\text{OBCO}) = \frac{1}{2} \text{ ar } (\text{OBCNO}) = \frac{\pi}{4} \times \frac{1}{2} = \pi/8$$

$$\text{Total Area} = \frac{\pi}{8} + \frac{\pi}{8} = \frac{\pi}{4} = \frac{\pi}{\lambda}$$

$$\text{Ans} = 4$$

6. **Ans (0)**

$$f(t) = \int_0^t e^x \sin(t-x) dx$$

$$f(t) = \int_0^t e^{t-x} \sin(x) dx$$

$$\Rightarrow f(t) = e^t \int_0^t e^{-x} \sin x dx$$

$$\Rightarrow f'(t) = e^t e^{-t} \sin t + \left( \int_0^t e^{-x} \sin x dx \right) e^t$$

$$\Rightarrow f'(t) = \sin t + f(t) \dots (1)$$

$$\Rightarrow f''(t) = \cos t + f'(t)$$

$$\Rightarrow f''(t) = \cos t + \sin t + f(t) \text{ {From (1)}}$$

$$f''(t) - f(t) = \sin t + \cos t$$

$$g(t) = \sin t + \cos t$$

$$\text{Range } [-\sqrt{2}, \sqrt{2}]$$

7. **Ans ( 15 )**

$$(2+t^3(t^2+6t+9))^{10}$$

↓

$$^{10}C_r \cdot 2^{10-r} \cdot t^{3r} \cdot (t+3)^{2r}$$

↓

$$^{2r}C_\lambda \cdot 3^{2r-\lambda} \cdot t^\lambda$$

$$\text{Now, } 3r = 8$$

$$r = 2, \lambda = 2$$

$$\therefore \text{coefficient of } t^8 \text{ is } ^{10}C_2 \cdot 2^8 \cdot ^4C_2 \cdot 3^2$$

$$\Rightarrow 5^1 \cdot 2^9 \cdot 3^5$$

$$a + b + c = 15$$

8. **Ans ( 4 )**

$$A + B + C = C + D + E = E + F + G = G + H + I = 13$$

$$\Rightarrow \frac{A+B+D+F+H+I}{X} + \frac{2(C+E+G)}{Y} = 52$$

Also,

$$A + B + C + \dots + I = 1 + 2 + 3 + \dots + 9$$

$$x + y = 45$$

$$x + 2y = 52$$

$$\text{—————}$$

$$-y = -7, \quad y = 7$$

$$\text{i.e. } C + E + G = 7$$

{1, 2, 4} - only one set possible

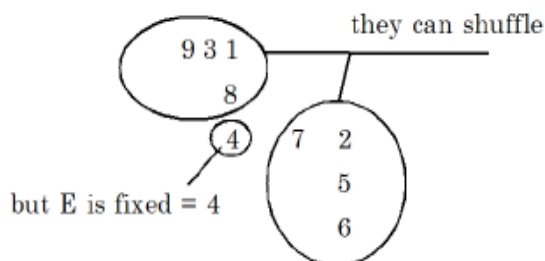
E can be 1, 2, 4

$E \neq 1$ , as 1 will come with 9 and 3

1 common goes with 8 and 4 then 4 goes with 7 and 2

$$\therefore E = 4$$

One possible Arrangement



9. **Ans ( 11 )**

$$\text{Put } z = \omega \text{ or } z = \omega^2.$$

10. **Ans ( 9 )**

$$\sigma^2 = \frac{20 \times 1 + 40 \times 2^2}{60} = 3$$

(Because mean of both samples is the same)